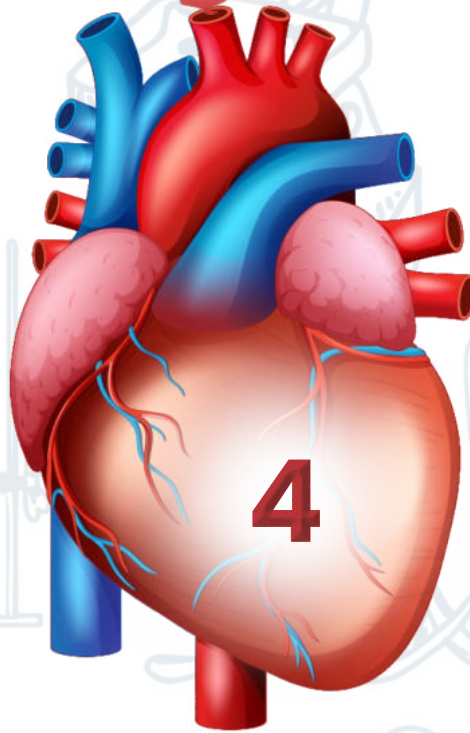
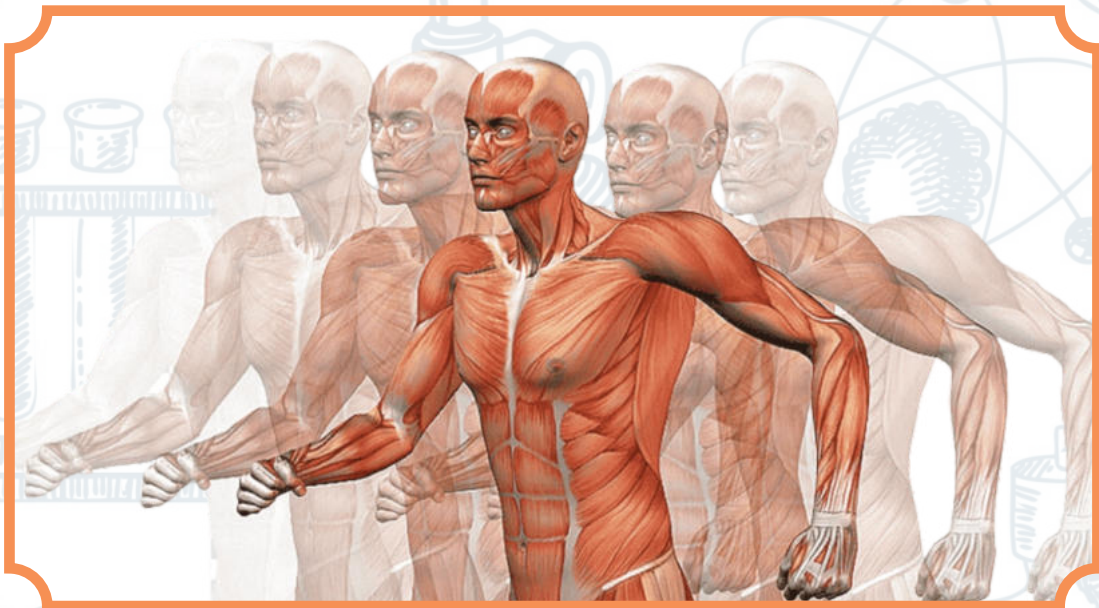


UNIT



BIOENERGETICS



BIOENERGETICS

All metabolic reactions involve energy transformations. So the quantitative study of energy transformation in biological system is called bioenergetics.

PHOTOSYNTHESIS

Chemically photosynthesis is a ; process in which CO_2 is reduced into glucose. Water acts as reducing agent which is oxidized into oxygen during this process.

Bio-energetically photosynthesis can be defined as an energy conversion process in which CO_2 and H_2O are transformed into energy rich molecule such as glucose.

The overall reaction of photosynthesis can be summarized as follows;



This process involves the interaction of sunlight, pigments, water and carbon dioxide.

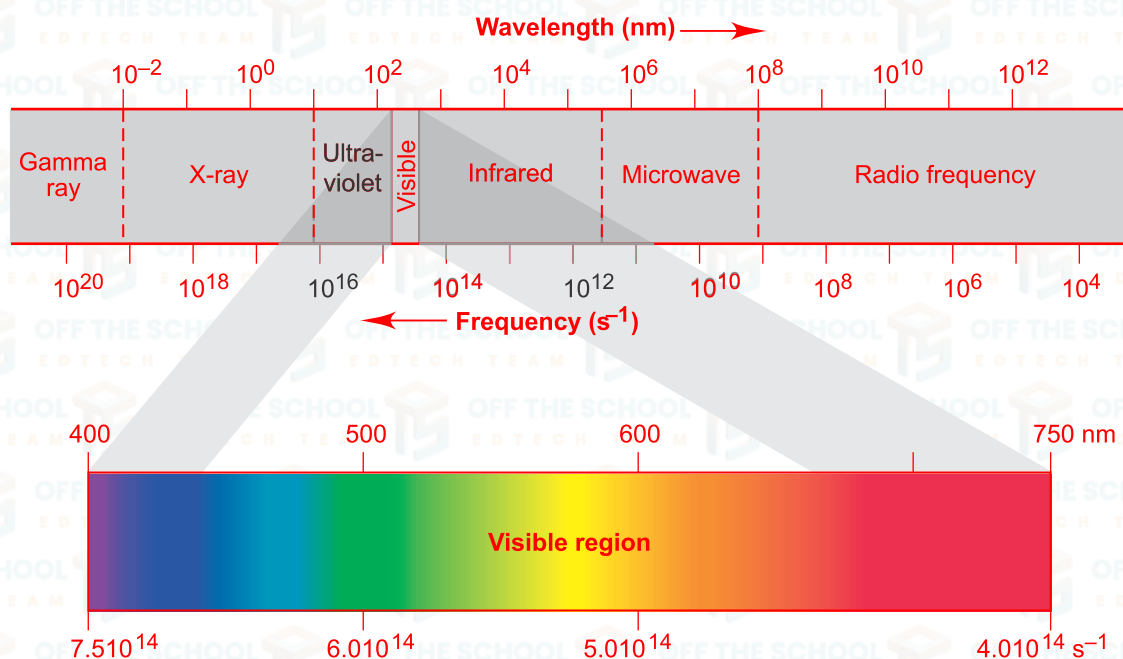
Role of Light

1) Nature of Light:

Sunlight is an electromagnetic form of energy.

2) Electromagnetic Spectrum:

The full range of electromagnetic radiation in the universe is called electromagnetic spectrum.

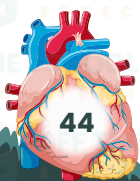


3) Visible Light:

Visible light is only a small part of the spectrum between 380 nm to 750 nm which is not only seen by naked eye but is also effective for the process of photosynthesis.

4) Effectiveness of Wavelength:

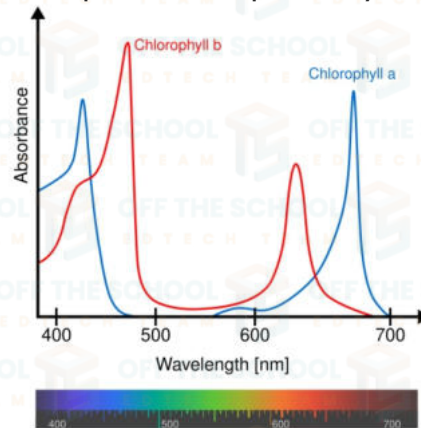
The effectiveness of a particular wavelength of light for the process of photosynthesis primarily depends upon its absorption in plant body. As different wavelengths (colours) of visible light are differently absorbed by various photosynthetic pigments, therefore, each wavelength has its own effectiveness for the process of photosynthesis.



5) Action Spectrum:

A graph which shows the effectiveness of different wavelength of light for the process of photosynthesis is called action spectrum.

- If a plant is illuminated in different colours of light one by one, the rate of photosynthesis is measured.
- Data obtained in this way is plotted in a graph, the rate of photosynthesis will be variable in different colours of light.
- Analysis of action spectrum indicates that blue (430 nm) and red (670 nm) wavelengths of light are the most effective for the process of photosynthesis.



Role of Photosynthetic Pigments

1) Pigment (Photosynthetic Pigment)

Pigment is any substance that absorbs light energy. All the wavelengths which are absorbed by a pigment are disappeared. A particular pigment shows only those wavelengths which are reflected back

2) Location

All the pigments that take part in photosynthesis are embedded in thylakoid membranes (grana lamellae) within chloroplasts.

3) Types

Plants have two major group of pigments i.e.

- Chlorophyll • Carotenoids

a) Chlorophyll

i) Absorbed Wavelengths

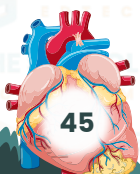
Chlorophylls absorb mainly violet, blue, orange and red wavelengths.

ii) Reflected Wavelengths

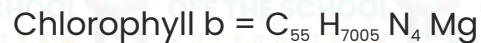
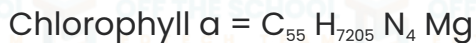
Green and yellow are least absorbed and are transmitted or reflected

iii) Types of Chlorophyll

- Two major types of chlorophyll are Chlorophyll-a and Chlorophyll-b
 - Chlorophyll-a: It is a bluish green pigment which is found in all photosynthetic organisms except photosynthetic bacteria
 - Chlorophyll-b: It is yellowish green pigment which is also found in all photosynthetic organisms except brown, red algae and photosynthetic bacteria
 - Other Types: Algae also have some other form of chlorophylls i.e., Chl-c, Chl-d and Chl-e while photosynthetic bacteria have yet another type of chlorophyll i.e., bacteriochlorophyll



iv) Molecular Formula



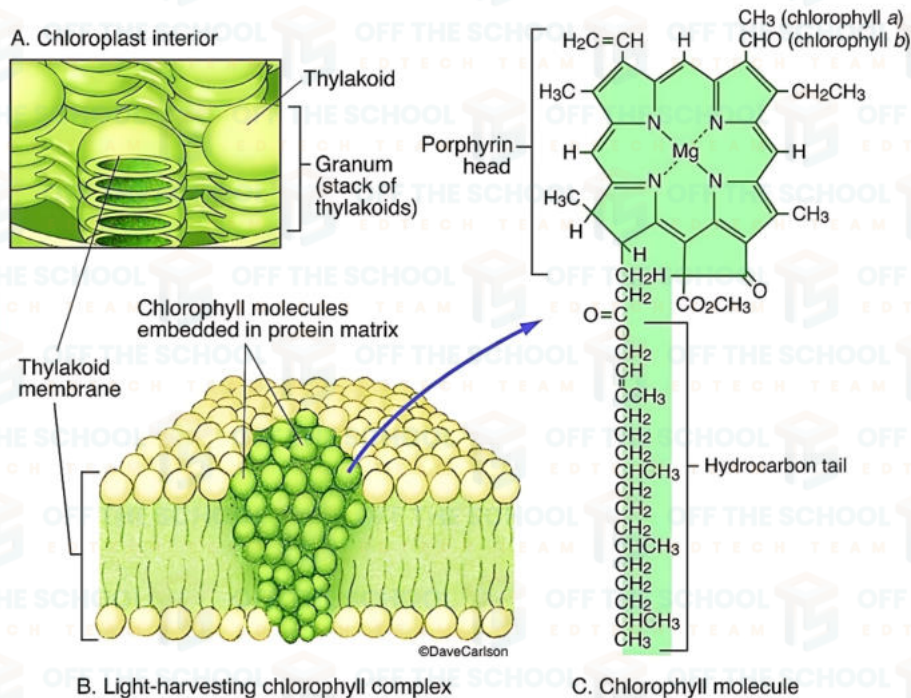
v) Structure of Chlorophyll

A molecule of chlorophyll consists of a head and two tails.

Head Structure

- The head is composed of a porphyrin ring with Mg in the centre.
- The porphyrin ring further consists of four pyrrole rings.
- Each pyrrole ring contains four carbons and one nitrogen atom.
- The nitrogen atoms of pyrrole rings interact with central Mg atom.
- The pyrrole rings also contain different groups around them.
- The only difference between chlorophyll-a and chlorophyll-b is that chlorophyll-a has methyl.

group ($-\text{CH}_3$) on 1 pyrrole ring whereas, chlorophyll-b has aldehyde group ($-\text{CHO}$) at this point



Nature

- The head of chlorophyll is hydrophilic in nature. Function
- It is exposed on the surface of thylakoid membrane. It is light absorbing part of chlorophyll.

The two side chains in the chlorophyll molecule are called tails.

Structure

- One side chain consists of a long chain phytol.
- The other side chain is methyl ester.

Carotenoids

- Carotenoids are terpenoid lipids, which are yellow, orange, red or brown pigments.
- They absorb light strongly in the blue-violet range.
- They are seen in leaves before leaf fall, present in some flowers and fruits.

ii) Types

There are two types of carotenoids:
carotenes and xanthophylls

a) Carotenes

- The carotenes are orange red pigments, composed of isoprenoid units
- They are found in all photosynthetic eukaryotes.
- The most widespread and important carotene is β (beta) carotene

b) Xanthophylls

- They are yellow in colour and are also composed of isoprenoid units.
- Lutein is widely distributed xanthophylls which is responsible for yellow colour of foliage in autumn

c) Role of Carotenoids

- The carotenoids act as accessory pigment along with chlorophyll-b as they absorb light energy and then transfer it to the chlorophyll-a
- They protect the chlorophyll-a from excess of light
- They also attract insects, birds and other animals for pollination and dispersal

Absorption Spectrum

A graph which shows the absorption of different colours of light by a particular pigment is called absorption spectrum

Measurement

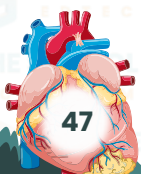
The absorption of different colours of light by a particular pigment can be determined by the help of spectrophotometer.

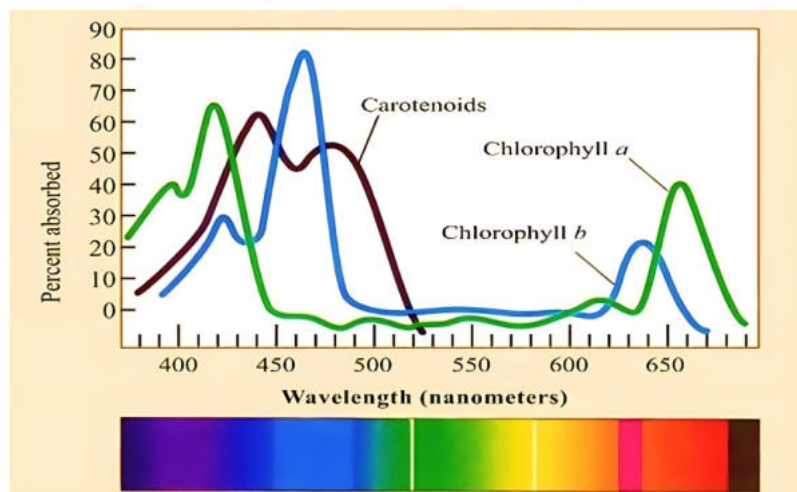
Conclusion

- The absorption spectra of different pigments indicate that they absorb different wavelengths of visible light

Absorption Spectrum of Photosynthetic Pigments

- The main photoreceptors are chlorophyll a and b and both show more absorption in violet blue (400 nm to 470 nm) and orange-red (630 nm to 660 nm) regions of the visible spectrum.
- On the other hand carotenoids show more absorption at 430 nm to 500 nm





Arrangements of Pigments (Photosystems)

The photosynthetic pigments are arranged in the form of clusters in thylakoid membranes. These clusters are called photosystems.

Advantage

Efficient absorption and utilization of light energy

Structure/ Components

Photosystems consists of two parts;

a) Antenna Complex

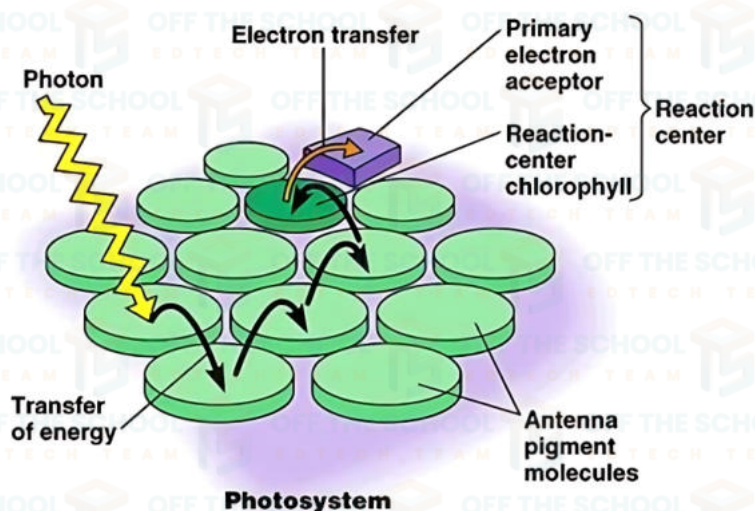
- It is peripheral part
- It contains accessory pigments i.e. chlorophyll-b and carotenoids

b) Reaction Centre

- It is the central part
- It contains chlorophyll-a (main pigment) only and associated proteins

Role of Water in Photosynthesis

Role of Water in Photosynthesis



Role of CO₂ in Photosynthesis

- Carbon dioxide acts as carbon source for the synthesis of organic compounds in photosynthesis. Carbon dioxide is utilized in the dark or light independent reaction (Calvin cycle) of photosynthesis.
- Land plants get carbon dioxide from atmosphere
- CO₂ with air enter the leaves through intercellular spaces called stomata which is dissolved in water and absorbed by mesophyll cells.
- Aquatic photosynthetic organisms use dissolved carbon dioxide, bicarbonates and carbonates present in water

Role of Water in Photosynthesis

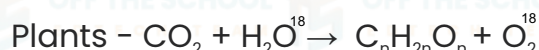
Water acts as hydrogen and electron donor in photosynthesis.

It replaces the electron lost by the P₆₈₀ in light reactions. 2H⁺ ions are taken up the NADP⁺ to form NADPH. The oxygen which is produced is released in atmosphere.

- Role of water as oxygen source in photosynthesis was first reported by Van Niel in 1930.
- He hypothesized that plants split water as a source of hydrogen, releasing oxygen as a byproduct.
- This observation was based on investigations of photosynthesis in bacteria that make carbohydrates from carbon dioxide, but do not release oxygen.
- Neils hypothesis was confirmed in 1940, when for the first time 18O in biological research was used.

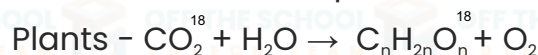
Experiment -1

- In first experiment water was made with 18 O
- The water tagged 18 O was added to an alga suspension.
- The oxygen, evolved during photosynthesis, was found to be radioactive. It was separated and identified.



Experiment-2

- In another experiment carbon dioxide with tagged 18 O was added.
- The oxygen evolved contained none of the isotopes.



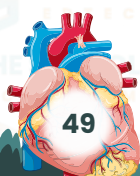
Conclusion

Thus the source of evolved oxygen was proved to be water.

Mechanism of photosynthesis

The process of photosynthesis has been divided into two phases.

- Light Dependent Phase / Light Phase / Light Reaction
- Light Independent Phase / Dark Phase / Dark Reaction



The Light-Dependent Reaction

1. Photolysis of water

- Light energy is used to breakdown water in a reaction known as **photolysis**; this produces hydrogen ions, electrons, and oxygen in the thylakoid lumen.
- Photosystem II contains a water-splitting enzyme which catalyses the breakdown (**photolysis**) of water by light:

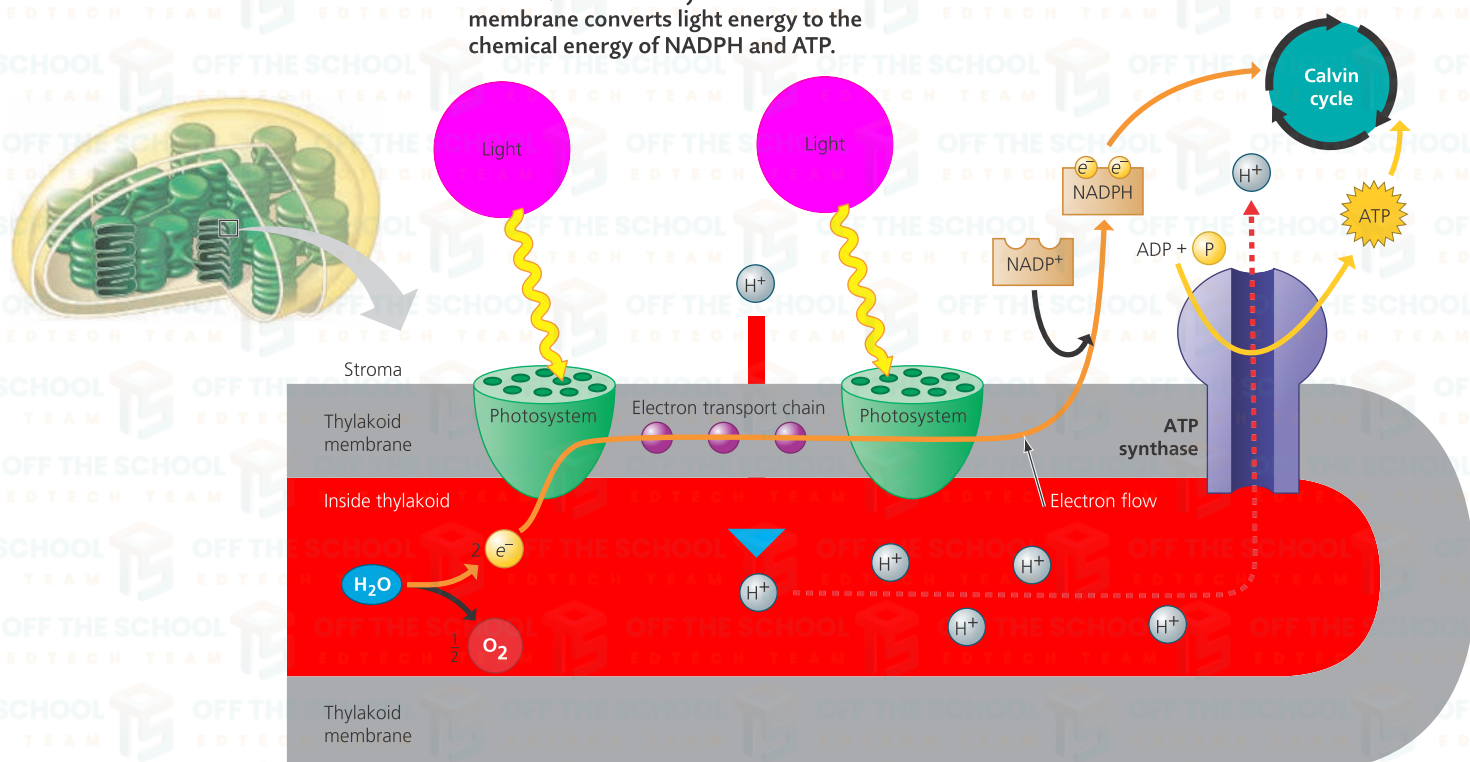


- As the excited electrons **leave** the primary pigment of photosystem II and are passed on to photosystem I, they are **replaced** by **electrons** from the **photolysis** of water

2. Electron transport chain

- When sunlight strikes **photosystem II (P680)** energy is absorbed by the chlorophyll molecules. The activated chlorophyll loses its two electrons and the positively charged chlorophyll molecule are left in the photosystem with a gap of two electrons.
- The high-energy electrons are captured by an electron acceptor called **Pheophytin** and then **Plastoquinone (PQ)**. The electrons pass along a series of electron acceptor molecules from one to another in the oxidation-reduction process from plastoquinone. These electron acceptors include cytochrome '**b**', **cytochrome 'f' (Cyt b/f)**, and **Plastocyanin**. This transfer of electrons constitutes an electron transport chain.
- The Plastocyanin acts as electron donor to Photosystem-I (P700), which passes the excited electron to a second electron transport chain. At the same time, light falls on photosystem I and activates its two electrons. Activated electrons are received by Ferredoxin which then transmit this electron to iron containing complex called **Ferredoxin**.

▼ How the thylakoid membrane converts light energy to the chemical energy of NADPH and ATP.



3. Reduction of NADP

- An enzyme called NADP reductase then transfer the electron from Ferridxin to to NADP.
- When electrons from the electron transport chain and hydrogen ions in the stroma combine with the NADP molecule,NADPH 2 is produced.

4. Photophosphorylation

- The excited electrons gradually release their energy as they pass through the electron transport chain.
- The released energy is used to actively transport protons (H^+ ions) across the thylakoid membrane.
- A '**proton pump**' transports the protons across the thylakoid membrane, from the stroma to the thylakoid lumen.
- The energy for this active transport comes from the excited electrons moving through the electron transport chain.
- This creates a proton gradient, with a high concentration of protons in the thylakoid lumen and a low concentration in the stroma.
- Protons then return to the stroma by facilitated diffusion through **ATP synthase** enzymes in a process known as **chemiosmosis**.
- This process provides the energy needed to **synthesis ATP**.



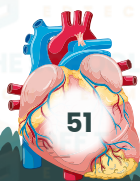
- The whole process is known as photophosphorylation.

Z-Scheme (Zigzag Scheme) of light reaction

In a non-cyclic electron transport system, electrons do not move in a cycle. Both the photosystems are involved and ATP and NADPH are produced which are used in dark reaction. Therefore, it is also called Z-Scheme (Zigzag Scheme) of light reaction. The ATP synthesis during this non-cyclic electron flow is called **Non-cyclic Photophosphorylation**.

Cyclic Photophosphorylation

- Cyclic photophosphorylation involves the use of only one photosystem (PS I) and does not involve the reduction of $NADP^+$
- When light is absorbed by Photosystem I, the excited electron may enter into an electron transport chain to produce ATP
- Following this, the de-energised electron returns to the photosystem, restoring its electron supply (hence: cyclic)
- As the electron returns to the photosystem, $NADP^+$ is not reduced and water is not needed to replenish the electron supply



LIGHT INDEPENDENT REACTION/ CALVIN BENSON CYCLE/ DARK REACTION

The cyclic series of reactions, catalyzed by enzymes, by which the carbon is fixed and reduced, resulting in the synthesis of sugar during the dark reactions of photosynthesis is called Calvin Cycle.

- Occurs daytime when ATP and NADPH are available
- Input ATP, NADPH and CO_2
- CO_2 is reduced using NADPH and ATP
- The product is a 3-C sugar called glyceraldehyde-3-phosphate (G_3P)
- Also called C_3 pathway and plants are called C_3 plants.

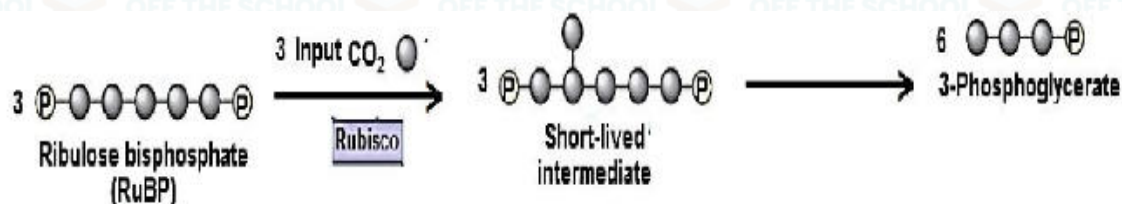
Phases

There are three phases of Calvin cycle:

- Carbon fixation
- Reduction
- Regeneration of CO_2 acceptor (RUBP)

a) Carbon fixation

- The CO_2 enters into the organic material (RUBP)
- The Calvin cycle begins when a molecule of CO_2 reacts with ribulose biphosphate (RUBP)
- This reaction is catalyzed by the enzyme ribulose biphosphate carboxylase, also known as Rubisco.
- The product of this reaction is a highly unstable, six carbon .
- It immediately breaks into two molecules of three - carbon compound called 3-phosphoglycerate (phosphoglyceric acid- 3PGA).



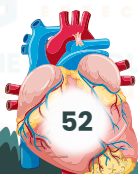
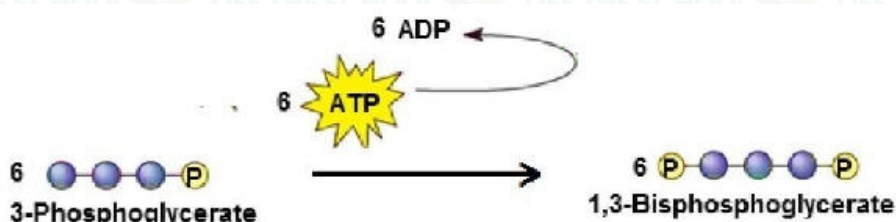
- The carbon that was part of CO_2 , become a part of an organic molecule. It means that carbon has been 'fixed'. Hence, it is called carbon fixation.
- The first stable compound of Calvin cycle is a 3-C compound (3PGA) hence Calvin cycle is also known as C_3 pathway.

b) Reduction

In this phase two steps are involved;

i) Formation of 1-3 bisphosphoglycerate

- Each molecule of (3-PGA) receives an additional phosphate from ATP • As a result 1,3-bisphosphoglycerate (1,3-bisPGA) is formed



A diagram illustrating the conversion of G3P sugar. On the left, a vertical chain of three red circles is labeled 'G3P sugar'. The bottom red circle is connected to a yellow circle labeled 'P'. A large black arrow points from this structure to a green hexagon on the right. Inside the hexagon, the text 'Glucose (and other organic compounds)' is written in white.

CELLULAR RESPIRATION

Cellular respiration is a series of complex oxidation-reduction reactions by which living cells obtain energy through the breakdown of organic matter.

Types of Cellular Respiration

There are two types of respirations:

aerobic respiration and anaerobic respiration.

- Aerobic respiration takes place in the presence of abundant atmospheric oxygen.
- whereas, anaerobic respiration occurs in the absence of oxygen.
- The initial breakdown of glucose in both aerobic and anaerobic respirations is same, in which it is broken down into two molecules of pyruvates.
- This common step of aerobic and anaerobic respirations is called glycolysis.

The Fate of Pyruvate

The pyruvates undergo in different respiratory pathways depending upon the availability of oxygen and the kind of organism.

i) When O_2 Present

- If oxygen is available, the further breakdown of pyruvates takes place aerobically and
- the final products are carbon dioxide and water • with the release of large amount of energy i.e., 36 ATPs (in eukaryotes) or 38 ATPs (in prokaryotes).

ii) When O_2 Absent

- If oxygen is absent, then the pyruvates are broken down anaerobically • and the final products are either lactic acid or ethanol and carbon dioxide • with release of very small amount of energy i.e., just 2 ATPs.

Mechanism of Anaerobic Respiration

Anaerobic respiration takes place in many microorganisms (bacteria, yeast), muscle cells of vertebrates and in the cells of higher plants.

- Anaerobic respiration is incomplete breakdown of glucose in the absence of oxygen. It is also known as fermentation.

- There are two pathways of anaerobic respiration depending upon the nature of final products I e., lactic acid fermentation and alcoholic fermentation.

1) Lactic Acid Fermentation

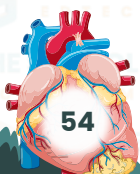
It consists of glycolysis followed by the reduction of pyruvate by NADH to lactic acid.

- The pathway operates anaerobically.
- Because after NADH transfers its electron to the pyruvate, it is to return and pick up more electrons during the earlier reaction of glycolysis.

Occurrence

Lactic acid fermentation occurs;

- in anaerobic bacteria and
- in the muscles of mammals as well as human during strenuous exercise when **oxygen** supply is exhausted.

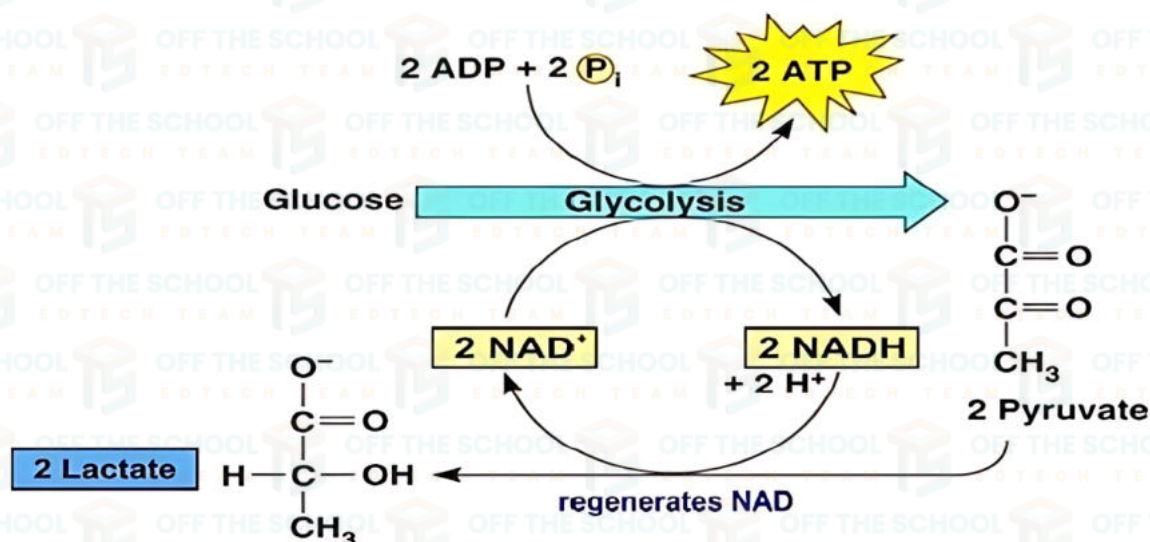


The accumulation of lactic acid causes muscles fatigue i.e., muscles become unable to contract and begin to ache.

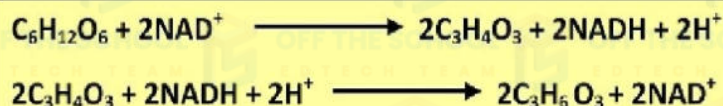
2) Alcoholic Fermentation

It consists of glycolysis followed by the decarboxylation of pyruvate to acetaldehyde then reduction of acetaldehyde by NADH to ethyl alcohol or ethanol.

- This pathway operates anaerobically
- Because after NADH transfers its electron to the acetaldehyde, it is "free" to return and pick up more electrons during the earlier reaction of glycolysis



- The overall equation can be represented as;



Occurrence

Alcoholic fermentation is found in yeast.

Mechanism of Aerobic Respiration

Aerobic Respiration

Aerobic respiration is a catabolic process which involves complete oxidative breakdown of organic food (especially glucose) into carbon dioxide and water with release of great deal of energy in the form of ATPs.

Phases of Aerobic Respiration

Aerobic respiration is completed in four phases:

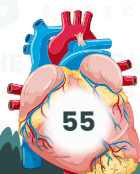
- Glycolysis,
- Oxidation of Pyruvates
- Krebs cycle
- Respiratory electron transport chain

Glycolysis

Break down of glucose into two pyruvate molecules Site It takes place in cytoplasm

Phases of Glycolysis

- It is completed in two phases i.e., preparatory phase and oxidative phase



a) Preparatory Phase

- Preparatory phase is an investment phase
- Two ATPs are consumed.
- Its end products are two molecules of G3P

b) Oxidative Phase

- Oxidative phase is pay off phase
- ATPs are produced through substrate level phosphorylation
- It produces NADH, which upon further oxidation in respiratory electron transport chain yields more ATPs.

a) Preparatory Phase**Step 1**

In the process, a phosphate group from ATP is transferred to glucose producing glucose 6-phosphate or G6P. One molecule of ATP is consumed during this phase.

Step 2

The enzyme isomerizes G6P into its isomer fructose 6-phosphate or F6P. Isomers have the same molecular formula as each other but different atomic arrangements.

Step 3

The enzyme uses another ATP molecule to transfer a phosphate group to F6P in order to form fructose 1,6-bisphosphate or FBP. Two ATP molecules have been used.

Step 4

The enzyme splits fructose 1,6-bisphosphate into a ketone and an aldehyde molecule. These sugars, dihydroxyacetone phosphate (DHAP) and glyceraldehyde 3-phosphate (G3P), are isomers of each other.

Step 5

DHAP converted into G3P converted (these isomers can inter-convert).

b) Oxidative Phase**Step 6**

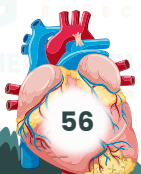
G3P will be converted into 1,3-bisphosphoglycerate (BPG). Next, Nicotinamide adenine dinucleotide (NAD^+) converted into NADH^+ & H^+ .

Step 7

This reaction yields two 3-phosphoglycerate (3 PGA) molecules and two ATP will be form from ADP.

Step 8

3 PGA will be converted into 2-phosphoglycerate (2 PGA) molecules.

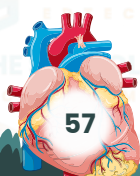
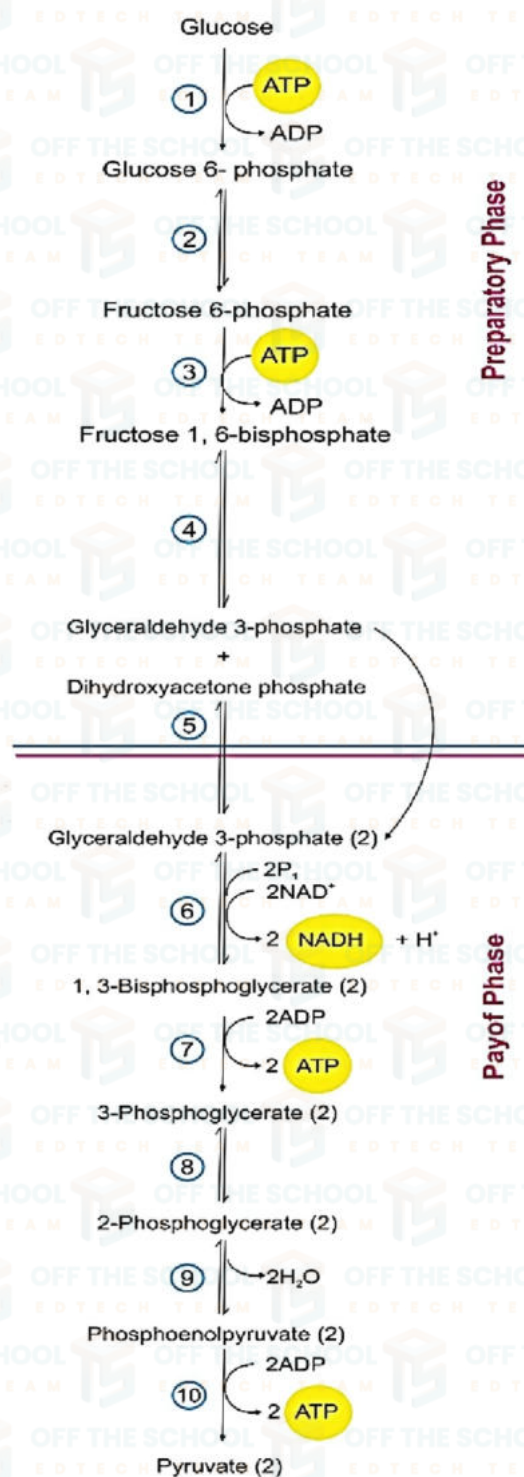


Step 9

A molecule of water will be removed from 2-phosphoglycerate to form phosphoenolpyruvate (PEP).

Step 10

A Phosphate is transferred from PEP to ADP to form ATP and pyruvate. This reaction yields two molecules of pyruvate and two ATP molecules.



Oxidation of Pyruvate

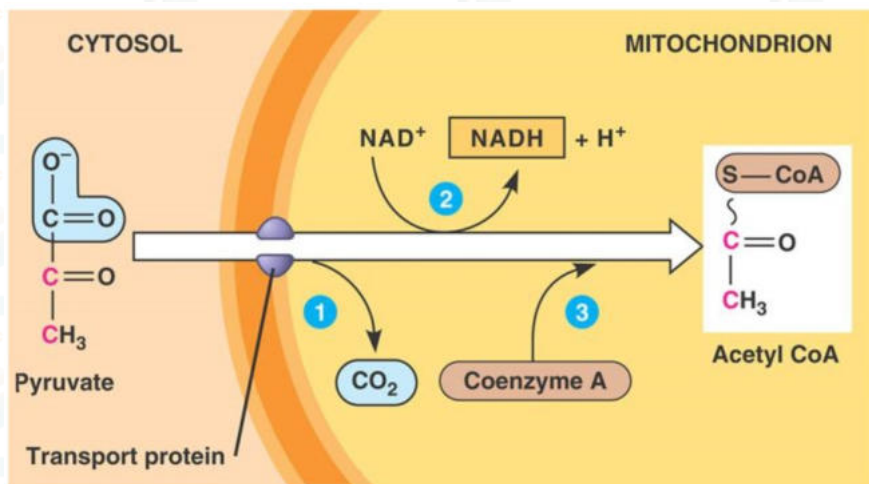
- Pyruvates are produced in cytosol
- Before Krebs cycle it is changed into Acetyl-Co-enzyme A

Entrance in Mitochondria

- Pyruvate is a charged molecule
- It must enter the mitochondrion via active transport with the help of the transport protein

Pyruvate Oxidation / Link Reaction

- On entering the mitochondria, pyruvates do not directly participate in Krebs cycle
- They undergo an intermediate phase, called oxidation of pyruvate or link reaction
- It links the pathway of aerobic respiration that occurs outside the mitochondria with occurs inside the mitochondria



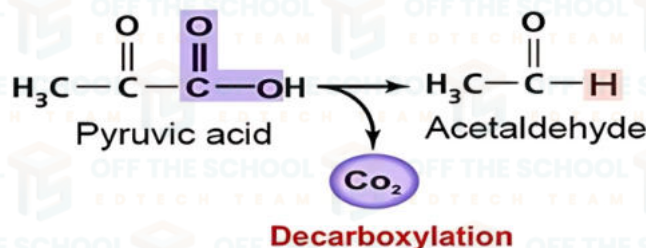
Steps of Pyruvate Oxidation

The oxidation of pyruvate takes place in 3 steps;

- Decarboxylation
- Dehydrogenation/ Oxidation
- Addition of Co-A

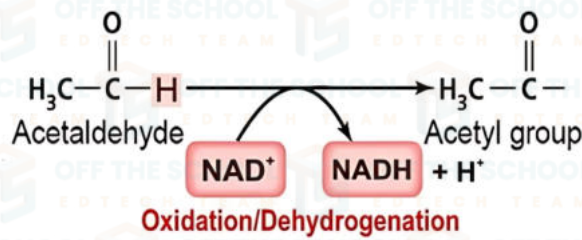
1. Decarboxylation

- One C-atom is removed from pyruvate in the form of CO_2
- This is called decarboxylation
- It results in formation of acetaldehyde



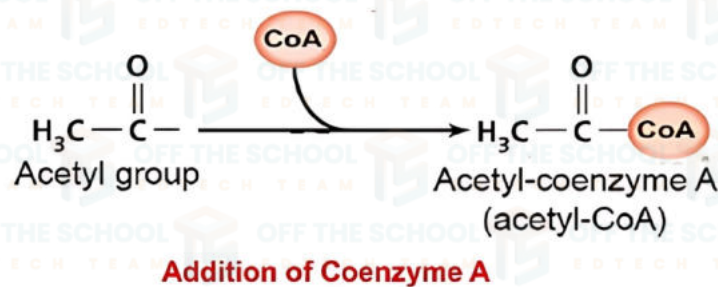
2. Dehydrogenation/Oxidation

- NAD^+ removes hydrogen from acetaldehyde
- As a result a 2C fragment acetyl and NADH are produced



3. Addition of Co-A

Acetyl group is combined with coenzyme-A to form acetyl CoA



Krebs Cycle

A cyclic series of chemical reactions in which Acetyl CoA is oxidized completely to CO_2 and ATP is generated while the hydrogen atoms are removed which reduces NAD and FAD to NADH_2 and FADH_2 respectively.

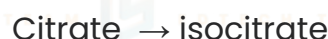
i- Formation of citrate

In the first step of the cycle, Acetyl CoA combines with pre-existing oxalo acetic acid (4-C) in the presence of a water molecule to form citric acid (6-C). CoA becomes free and is ready to react with another acetyl group.



ii- Conversion of citrate to isocitrate

Citrate is converted to isocitrate



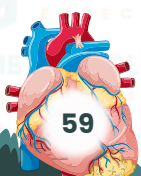
iii- Formation of -ketoglutarate

Iso-citrate is oxidized to α -ketoglutarate (5-C). In this step, one carbon of isocitrate is oxidized to carbon dioxide and hydrogen is removed which is picked up by NAD reducing into NADH_2 .

One carbon of α -Ketoglutarate is oxidized into carbon dioxide. The CO_2 is released from the cycle. This is the second CO_2 molecule produced in the Krebs cycle.

iv- Formation of succinyl CoA

The two carbons of the acetyl group which was entered into the Krebs cycle are oxidized into two molecules of carbon dioxide. Hydrogen atoms are released which are accepted by oxidized NAD. By accepting hydrogen atoms NAD is reduced to NADH. The α -Ketoglutarate is converted into succinyl group (4-C). CoA reacts with succinyl group forming succinyl CoA. oxaloacetate is now able to react with another acetyl Co-A and continue the cycle.



v- Formation of succinate acid

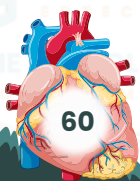
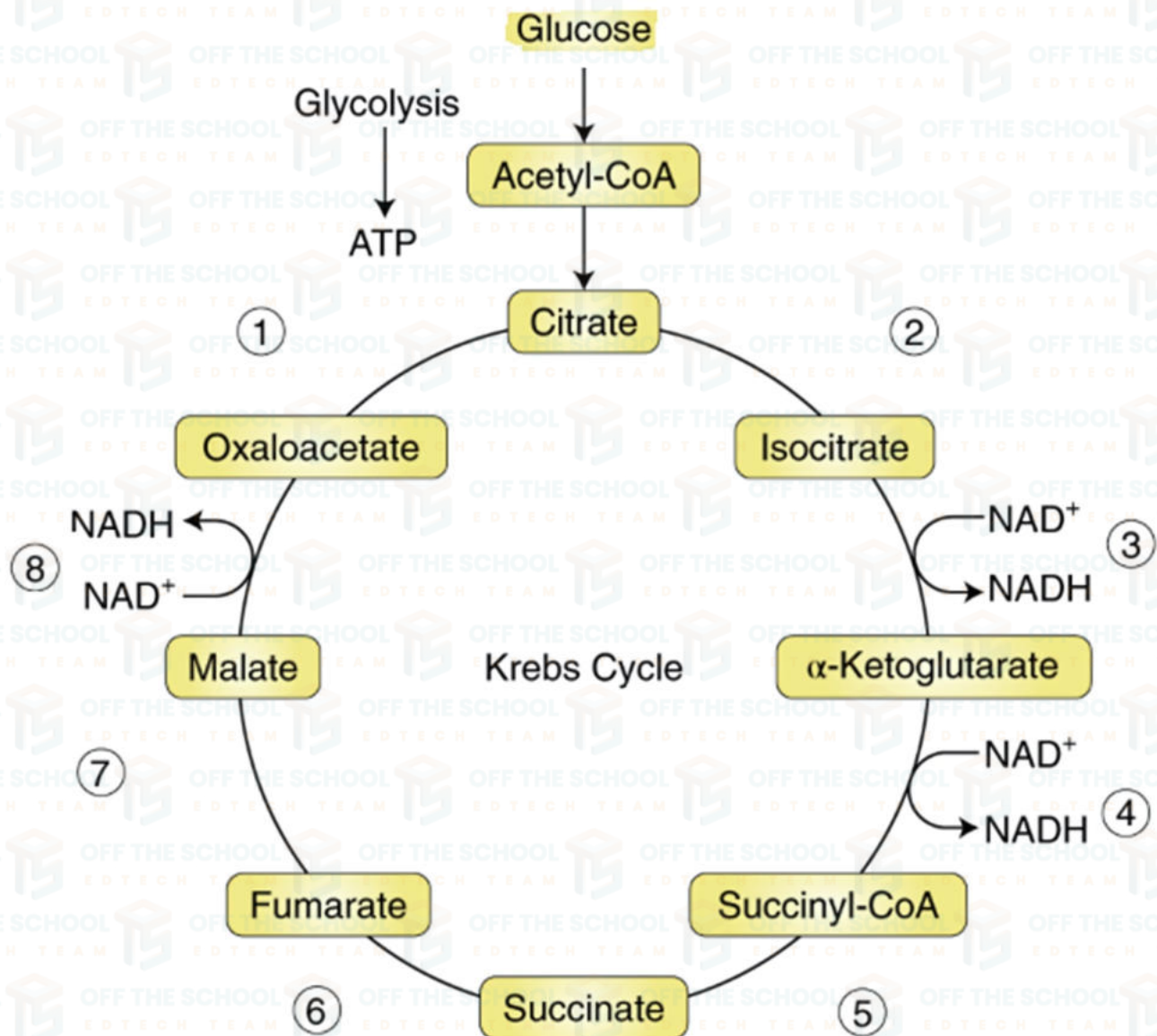
In the next step, succinyl CoA is converted to succinate Acid (4-C) and CoA is released. Some of the energy produced in the oxidation is used in the synthesis of ATP. The energy of the substrate used in the generation of ATP is called substrate-level phosphorylation.

vi- Oxidation of succinate to fumarate and malate

Succinate is oxidized to fumarate (4-C). Coenzyme FAD (Flavin adenine dinucleotide) is reduced in the reaction. Fumarate is converted to malate (4-C).

vii. Oxidation of malate into oxaloacetate

In the last step, malate oxidized to oxaloacetate. A molecule of NADH is produced during this step. The oxaloacetate is now able to react with another acetyl C0-A and continue the cycle.



Electron Transport Chain

- After Krebs cycle most of the energy of glucose is in the form of NADH and FADH₂
- These two molecules enter into the electron transport chain.
- NADH and FADH₂ are oxidized and their electrons are passed along a series of oxidation reduction reaction to the final acceptor i.e., molecular oxygen

Components of ETC

The components of electron transport chain include:

1. NADH- dehydrogenase complex (I)
2. FADH-dehydrogenase complex (II)
3. Coenzyme Q 4. Cytochrome reductase complex (III)
5. Cytochrome-c 6. Cytochrome oxidase complex (IV)

Passage of electron flow**i) NADH- dehydrogenase complex (I)**

- NADH is oxidized when it reacts with NADH- dehydrogenase complex (I)

ii) Co-enzyme Q

- Electrons now move to the co-enzyme Q.
- If FADH₂ is to be oxidized through ETC, it also hands over its electrons to coenzyme Q, via FADH dehydrogenase complex (II)

iii) Cytochrome reductase complex (III)

- The flowing electrons from coenzyme Q are now transferred to cytochrome reductase complex (III)
- It hands over its electron to cytochrome c.

iv) Cytochrome c

- Like co-enzyme Q, cytochrome c is also mobile carrier of electrons.

v) cytochrome oxidase complex

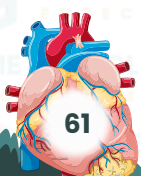
- Cytochrome c delivers the electrons to cytochrome oxidase complex (IV)
- Finally, the electrons are transferred to oxygen. • The oxygen is the ultimate acceptor of electrons. It becomes reactive.
- Each oxygen atom also picks up a pair of hydrogen ions from the aqueous solution forming water

Released Energy & Proton Pumping

- Energy released during passage of electrons from one carrier to the next is used to pump protons (H⁺) from the mitochondrial matrix to the inter membrane space.
- There are three such sites, corresponding to three enzymes present in the electron transport chain a) NADH-dehydrogenase complex (I) b) Cytochrome reductase complex (III) c) Cytochrome oxidase complex (IV)

The ATP Synthetase

- The energy of the proton gradient is used in attaching a phosphate to ADP to produce ATP



ATP Yield

- Oxidation of one molecule of NADH_2 produces 3 ATP molecules
- Oxidation of FADH_2 forms 2 ATP molecules

Substrate Level Phosphorylation

The addition of inorganic phosphate to any organic molecule is called phosphorylation but, when phosphate is enzymatically transferred from an organic substrates molecule it is called substrate level phosphorylation.

Substrate level phosphorylation accounts for only a small percentage of the ATP that a cell generates. It occurs at three occasions during aerobic respiration.

a) In Glycolysis

- i) 1,3-bisphosphoglycerate is converted into 3-phosphoglycerate (7th reaction)
- ii) Phosphoenol pyruvate is converted into pyruvate (10th reaction)

b) In Photosynthesis

- In the Calvin cycle of photosynthesis, G3P molecules are converted into glucose phosphate within the chloroplast.
- Glucose phosphate is then converted to glucose, fructose, sucrose and starch.

Cellular Respiration of Fats and Proteins

2) Fats as Energy Source

When a fat is used as an energy source, it breaks down to glycerol and three fatty acids.

- Glycerol is converted to G3P, a metabolite in glycolysis
- The fatty acids are converted to acetyl-CoA, which enters the Krebs cycle. An 18-carbon fatty acid results in nine acetyl-CoA molecules.

3) Proteins as Energy Source

The hydrolysis of proteins results in amino acids whose R-group size determines whether the carbon chain is oxidized in glycolysis or the Krebs cycle.

- The carbon chain is produced in the liver when an amino acid undergoes deamination, i.e., the removal of the amino group.
- The amino group becomes ammonia (NH_3), which enters the urea cycle and becomes part of urea.

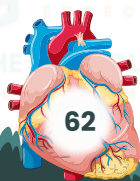
Photorespiration

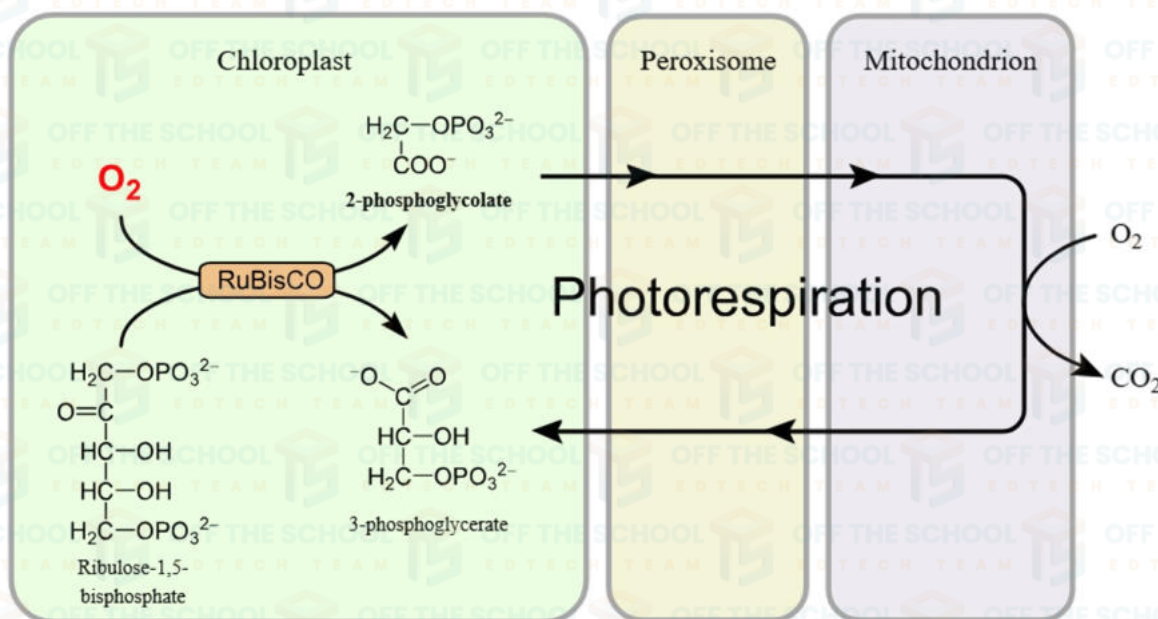
"Photorespiration is the process in which oxygen combines with ribulose biphosphate (RuBP) in the presence of sunlight and CO_2 is produced. The process is called photorespiration because in the presence of light (photo), oxygen is taken up and CO_2 is evolved (respiration)."

"Or"

"Photorespiration occurs in many higher plants by which they take up oxygen in the light and give out some carbon dioxide, contrary to the general pattern of photosynthesis."

In dry and hot weather plants close up stomata to conserve water. In such condition CO_2 cannot enter the leaf and O_2 cannot leave it. Dry and hot conditions are usually accompanied by intense sunlight therefore light reaction occurs with a maximum rate which results in maximum use of CO_2 . Since the concentration of CO_2 lowers down in the leaf and photorespiration proceeds.



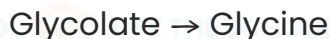


Steps of photorespiration:

1- Oxygen combines with RuBP (present in the stroma of chloroplast) and a compound called Glycolate is produced.



2. Glycolate is converted into glycine (simplest amino acid) in the peroxisome.



3- Glycine is transported to mitochondria where it is converted into serine and a molecule of CO_2 is produced.



Disadvantages of Photorespiration

Photorespiration is just the reverse of photosynthesis hampering the fixation of CO_2 , photosynthesis.

The process wastes energy and does nothing to serve the needs of the plant.

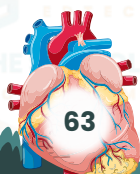
Effect of temperature on the activities of RuBisCO

- On hot dry days when plant begins to secrete abscisic acid which causes closing of stomata to prevent excess water loss.
- If the plant continues CO_2 fixation in photosynthesis when its stomata are closed, the CO_2 will be used up and the O_2 released from photosynthesis
- In this way, ratio of O_2 in the leaf will increase relative to CO_2 concentrations.
- Photorespiration starts when the CO_2 levels inside a leaf become low.

C 4 Photosynthesis

Some plants which grow in tropical climate have an adaptation to the problem of photorespiration.

- They have an additional metabolic pathway in light independent phase of photosynthesis beside Calvin cycle.
- This metabolic pathway is called Hatch-Slack cycle of C_4 pathway. Mechanism



i) CO₂ Fixation

In C₄ pathway phosphoenol pyruvate (PEP) carboxylase is used instead of RuBisCO to fix CO₂ to phosphoenol pyruvate (a C₃ molecule), and the result is oxaloacetate, a C₄ molecule.

- It takes place in cytoplasm of mesophyll cells.
- As the first product of CO₂ fixation is a 4-carbon compound oxaloacetate, so the plants are called C₄ plants e.g maize, sugarcane, sorghum, etc.

ii) Formation of Malate

- Oxaloacetate is then transported to the chloroplasts of mesophyll cells.
- It is then converted to another 4-C compound, the malate, with the help of NADH, produced in the photochemical phase. Some of the pyruvate generated in the bundle sheath cells re-enters mesophyll cells and regenerates phosphoenol pyruvate (PEP) by consuming one ATP.

iii) Formation of Pyruvate

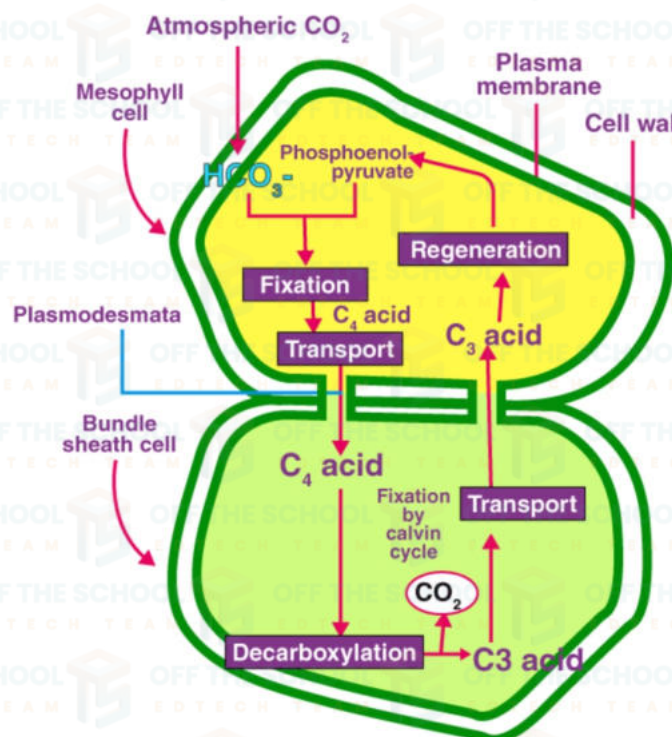
- The malate is then transported to the chloroplasts of bundle sheath cells.
- Here malate is converted to pyruvate (C₃) with the release of CO₂. Thus concentration of CO₂ increases in the bundle sheath cells.

iv) Calvin Cycle

- Bundle sheath cells contain enzymes of Calvin cycle.
- Because of high concentration of CO₂, RubisCO participates in Calvin cycle and not in photorespiration.
- Sugar formed in Calvin cycle is transported into the phloem.

v) Regeneration of PEP

Some of the pyruvate generated in the bundle sheath cells re-enters mesophyll cells and regenerates phosphoenol pyruvate (PEP) by consuming one ATP.



CAM pathway

CAM pathway is adapted in plants to perform photosynthesis under stress. The CAM pathway reduces photorespiration.

In CAM plants stomata are open at night and they absorb carbon dioxide at night to reduce water loss during the daytime. The process has the following steps:

- The first step in carbon dioxide fixation is the combination of CO_2 with PEP (phosphoenolpyruvate) to form 4 carbon oxaloacetate (same as C 4 plants) in the chloroplast of mesophyll cells. The reaction is catalysed by PEPcarboxylase. This occurs at night.
- Oxaloacetate is converted to malate and other C 4 acids. Malate is stored in vacuoles at night.
- During the daytime, stomata remain closed, so there is no gas exchange. Malate is transported out of the vacuole and CO_2 is released by the process of decarboxylation.
- This CO_2 finally enters the Calvin cycle and carbon fixation completes.
- The CO_2 which gets accumulated around RuBisCO increases the efficiency of the photosynthesis process and minimizes photorespiration.

